T03: Nonlinear Mathematical Derivation

Furuta Rotary Inverted Pendulum

Modern Control Course Project

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Abstract

This report fixes an unambiguous two-degree-of-freedom Furuta-pendulum coordinate system, derives its full mechanical dynamics with three pendulum principal inertias, and supplies executable Python and MATLAB-compatible interfaces. The model input is effective mechanical torque about the base axis. Viscous joint damping is included; electrical dynamics and non-viscous effects are explicitly deferred. An independent symbolic Euler–Lagrange construction agrees exactly with the separately coded compact model. All blocking validation tests V01–V09 pass, including mass-matrix positivity, energy balance, limiting cases, randomized algebra, and conservative and damped simulations. Numerical values used here are synthetic verification data, not measurements of the final apparatus.

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1 Scope and requirements traceability

The official course brief requires a nonlinear equation-of-motion derivation, explicit assumptions and parameters, consistent notation, and a model suitable for subsequent linearization and simulation. The T03 task cards divide that work into conventions, energies, nonlinear equations, actuator/friction assumptions, and independent validation. Table 1 maps those outputs to this package.

Table 1: T03 task-card traceability.

Task	Required output	Evidence in this package
T03.01	Coordinates and parameter meanings	Sections 2–8, Figure 1
T03.02	Kinematics and energy expressions	Sections 3–4
T03.03	Euler–Lagrange equations and acceleration form	Sections 5–6
T03.04	Motor, transmission, and friction assumptions	Section 7
T03.05	Independent verification record	Section 10 and `results/`

The source PDFs and workbook are requirements evidence only; no instruction embedded in them is treated as authorization to alter external files. The master tracker is therefore left unchanged. Recommended status updates are listed in `HANDOFF.md`.

2 Geometry, coordinates, and sign conventions

Let the inertial frame have vertical unit vector $\mathbf{e}_z$. The arm rotates in the horizontal plane. Its rotating radial and tangential basis is

$$\mathbf{e}_r = \begin{bmatrix}\cos\theta_0 & \sin\theta_0 & 0\end{bmatrix}^T, \qquad \mathbf{e}_\theta = \begin{bmatrix}-\sin\theta_0 & \cos\theta_0 & 0\end{bmatrix}^T, \tag{1}$$

$$\dot{\mathbf{e}}_r = \dot{\theta}_0 \mathbf{e}_\theta, \qquad \dot{\mathbf{e}}_\theta = -\dot{\theta}_0 \mathbf{e}_r. \tag{2}$$

The arm coordinate θ_0 is positive counterclockwise when viewed from $+\mathbf{e}_z$. The pendulum hinge axis is radial, $\mathbf{e}_r$, so the pendulum moves in the $\mathbf{e}_\theta$–$\mathbf{e}_z$ plane. The pendulum coordinate satisfies

$$\theta_1 = 0 \quad \text{upright}, \qquad \theta_1 = \pi \quad \text{downward}, \tag{3}$$

and increases from $+\mathbf{e}_z$ toward $+\mathbf{e}_\theta$. The generalized coordinates, state, and input are

$$\mathbf{q} = \begin{bmatrix}\theta_0 \\ \theta_1\end{bmatrix}, \qquad \mathbf{x} = \begin{bmatrix}\theta_0 & \theta_1 & \dot{\theta}_0 & \dot{\theta}_1\end{bmatrix}^T, \qquad \mathbf{u} = \begin{bmatrix}\tau \\ 0\end{bmatrix}. \tag{4}$$

Here τ is effective mechanical torque applied to the arm about $+\mathbf{e}_z$.

The arm length L_0 is the distance from the base axis to the pendulum pivot P. The total physical pendulum length is denoted L_p, while l_c is the pivot-to-centre-of-mass distance. They must not be interchanged. This distinction resolves the course brief's ambiguous l_1 label at the model boundary.

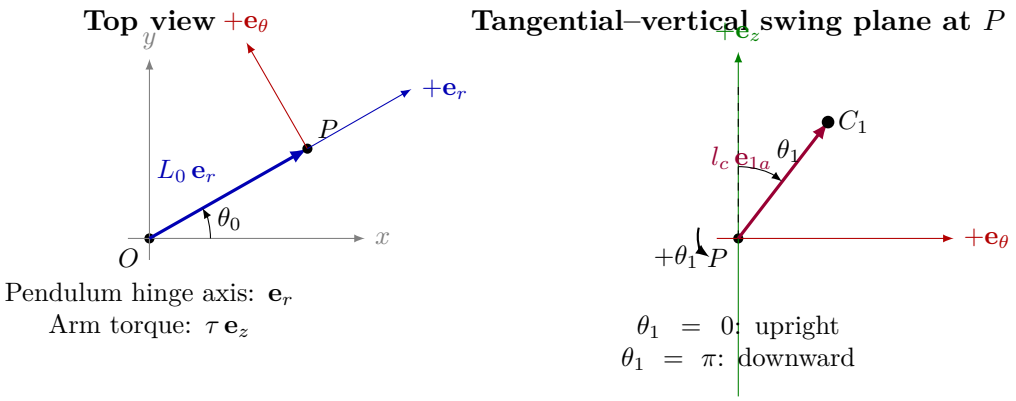


Figure 1: Adopted coordinate convention. The pendulum swings in the $\mathbf{e}_\theta$–$\mathbf{e}_z$ plane about the radial hinge axis $\mathbf{e}_r$.

2.1 Model assumptions

1. The base, arm, pendulum, and joints are rigid; the base is fixed.
2. The pendulum centre of mass lies on its longitudinal model axis.
3. The selected pendulum body axes are principal axes, so products of inertia vanish. CAD products of inertia require an explicit extension.
4. The arm and pendulum hinge axes are orthogonal and intersect at P.
5. Gravity is uniform and points in $-\mathbf{e}_z$.
6. Joint losses are linear viscous damping only. Coulomb friction, backlash, flexibility, saturation, cable forces, and encoder dynamics are excluded.
7. J_0 is the effective rotating base/arm inertia about $\mathbf{e}_z$, excluding the pendulum. Reflected motor inertia may be included once, but never both in J_0 and again as a separate term.
8. The T03 input is mechanical torque. Voltage is not treated as torque.

3 Kinematics

The arm-end pivot and pendulum centre-of-mass positions are

$$\mathbf{r}_P = L_0\mathbf{e}_r, \tag{5}$$

$$\mathbf{r}_{C_1} = L_0\mathbf{e}_r + l_c\left(\sin\theta_1\mathbf{e}_\theta + \cos\theta_1\mathbf{e}_z\right). \tag{6}$$

Differentiation using the rotating-basis identities gives

$$\mathbf{v}_{C_1} = -l_c\sin\theta_1\dot{\theta}_0\mathbf{e}_r + \left(L_0\dot{\theta}_0 + l_c\cos\theta_1\dot{\theta}_1\right)\mathbf{e}_\theta - l_c\sin\theta_1\dot{\theta}_1\mathbf{e}_z, \tag{7}$$

and therefore

$$\|\mathbf{v}_{C_1}\|^2 = \left(L_0^2 + l_c^2\sin^2\theta_1\right)\dot{\theta}_0^2 + 2L_0l_c\cos\theta_1\dot{\theta}_0\dot{\theta}_1 + l_c^2\dot{\theta}_1^2. \tag{8}$$

Choose pendulum body axes

$$\mathbf{e}_{1r} = \mathbf{e}_r, \tag{9}$$

$$\mathbf{e}_{1t} = \cos\theta_1\mathbf{e}_\theta - \sin\theta_1\mathbf{e}_z, \tag{10}$$

$$\mathbf{e}_{1a} = \sin\theta_1\mathbf{e}_\theta + \cos\theta_1\mathbf{e}_z, \tag{11}$$

where $\mathbf{e}_{1a}$ is longitudinal. The angular velocity resolved in these principal axes is

$$\boldsymbol{\omega}_1 = -\dot{\theta}_1 \mathbf{e}_{1r} - \dot{\theta}_0 \sin\theta_1 \mathbf{e}_{1t} + \dot{\theta}_0 \cos\theta_1 \mathbf{e}_{1a}. \tag{12}$$

The sign of the radial component follows from the chosen positive θ_1; its sign does not affect the diagonal rotational kinetic energy.

4 Energy and dissipation

Let J_{1r}, J_{1t}, and J_{1a} denote the pendulum COM principal inertias about $\mathbf{e}_{1r}$, $\mathbf{e}_{1t}$, and $\mathbf{e}_{1a}$. The three kinetic-energy contributions are

$$T_0 = \frac{1}{2} J_0 \dot{\theta}_0^2, \tag{13}$$

$$T_{1,\mathrm{tr}} = \frac{1}{2} m_1 \left[\left(L_0^2 + l_c^2 \sin^2\theta_1 \right) \dot{\theta}_0^2 + 2 L_0 l_c \cos\theta_1 \dot{\theta}_0 \dot{\theta}_1 + l_c^2 \dot{\theta}_1^2 \right], \tag{14}$$

$$T_{1,\mathrm{rot}} = \frac{1}{2} \left[J_{1r} \dot{\theta}_1^2 + J_{1t} \dot{\theta}_0^2 \sin^2\theta_1 + J_{1a} \dot{\theta}_0^2 \cos^2\theta_1 \right]. \tag{15}$$

Thus $T = T_0 + T_{1,\mathrm{tr}} + T_{1,\mathrm{rot}}$. With zero potential at the downward configuration,

$$V = m_1 g l_c (1 + \cos\theta_1). \tag{16}$$

The Rayleigh dissipation function is

$$\mathcal{R} = \frac{1}{2} b_0 \dot{\theta}_0^2 + \frac{1}{2} b_1 \dot{\theta}_1^2. \tag{17}$$

For compactness define

$$A(\theta_1) = J_0 + m_1 L_0^2 + (m_1 l_c^2 + J_{1t}) \sin^2\theta_1 + J_{1a} \cos^2\theta_1, \tag{18}$$

$$D = J_{1r} + m_1 l_c^2, \qquad B = m_1 L_0 l_c, \tag{19}$$

$$\delta = m_1 l_c^2 + J_{1t} - J_{1a}. \tag{20}$$

Then

$$T = \frac{1}{2} A \dot{\theta}_0^2 + B \cos\theta_1 \dot{\theta}_0 \dot{\theta}_1 + \frac{1}{2} D \dot{\theta}_1^2. \tag{21}$$

5 Euler–Lagrange derivation

With $\mathcal{L} = T - V$, the dissipative Euler–Lagrange equation is

$$\frac{d}{dt} \frac{\partial T}{\partial \dot{q}_i} - \frac{\partial T}{\partial q_i} + \frac{\partial V}{\partial q_i} + \frac{\partial \mathcal{R}}{\partial \dot{q}_i} = Q_i. \tag{22}$$

The required derivatives are shown before the terms are collected.

For $q_1 = \theta_0$,

$$\frac{\partial T}{\partial \dot{\theta}_0} = A \dot{\theta}_0 + B \cos\theta_1 \dot{\theta}_1, \tag{23}$$

$$\frac{d}{dt} \frac{\partial T}{\partial \dot{\theta}_0} = A \ddot{\theta}_0 + B \cos\theta_1 \ddot{\theta}_1 + 2\delta \sin\theta_1 \cos\theta_1 \dot{\theta}_0 \dot{\theta}_1 - B \sin\theta_1 \dot{\theta}_1^2, \tag{24}$$

$$\frac{\partial T}{\partial \theta_0} = 0, \qquad \frac{\partial V}{\partial \theta_0} = 0, \qquad \frac{\partial \mathcal{R}}{\partial \dot{\theta}_0} = b_0 \dot{\theta}_0. \tag{25}$$

For $q_2 = \theta_1$,

$$\frac{\partial T}{\partial \dot{\theta}_1} = B\cos\theta_1\dot{\theta}_0 + D\dot{\theta}_1, \tag{26}$$

$$\frac{d}{dt}\frac{\partial T}{\partial \dot{\theta}_1} = B\cos\theta_1\ddot{\theta}_0 - B\sin\theta_1\dot{\theta}_0\dot{\theta}_1 + D\ddot{\theta}_1, \tag{27}$$

$$\frac{\partial T}{\partial \theta_1} = \delta\sin\theta_1\cos\theta_1\dot{\theta}_0^2 - B\sin\theta_1\dot{\theta}_0\dot{\theta}_1, \tag{28}$$

$$\frac{\partial V}{\partial \theta_1} = -m_1 g l_c \sin\theta_1, \qquad \frac{\partial \mathcal{R}}{\partial \dot{\theta}_1} = b_1\dot{\theta}_1. \tag{29}$$

The two scalar equations are therefore

$$A\ddot{\theta}_0 + B\cos\theta_1\ddot{\theta}_1 + 2\delta\sin\theta_1\cos\theta_1\dot{\theta}_0\dot{\theta}_1 - B\sin\theta_1\dot{\theta}_1^2 + b_0\dot{\theta}_0 = \tau, \tag{30}$$

$$B\cos\theta_1\ddot{\theta}_0 + D\ddot{\theta}_1 - \delta\sin\theta_1\cos\theta_1\dot{\theta}_0^2 - m_1 g l_c\sin\theta_1 + b_1\dot{\theta}_1 = 0. \tag{31}$$

6 Compact nonlinear and state forms

Equations (30)–(31) are

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{c}(\mathbf{q},\dot{\mathbf{q}}) + \mathbf{g}(\mathbf{q}) + \mathbf{d}(\dot{\mathbf{q}}) = \mathbf{u}, \tag{32}$$

where

$$\mathbf{M} = \begin{bmatrix} A & B\cos\theta_1 \\ B\cos\theta_1 & D \end{bmatrix}, \tag{33}$$

$$\mathbf{c} = \begin{bmatrix} 2\delta\sin\theta_1\cos\theta_1\dot{\theta}_0\dot{\theta}_1 - B\sin\theta_1\dot{\theta}_1^2 \\ -\delta\sin\theta_1\cos\theta_1\dot{\theta}_0^2 \end{bmatrix}, \tag{34}$$

$$\mathbf{g} = \begin{bmatrix} 0 \\ -m_1 g l_c\sin\theta_1 \end{bmatrix}, \qquad \mathbf{d} = \begin{bmatrix} b_0\dot{\theta}_0 \\ b_1\dot{\theta}_1 \end{bmatrix}. \tag{35}$$

The acceleration is evaluated numerically by solving

$$\mathbf{M}\ddot{\mathbf{q}} = \mathbf{u} - \mathbf{c} - \mathbf{g} - \mathbf{d}, \tag{36}$$

never by explicitly forming $\mathbf{M}^{-1}$. The first-order nonlinear plant is

$$\dot{\mathbf{x}} = \begin{bmatrix} \dot{\theta}_0 \\ \dot{\theta}_1 \\ \mathbf{M}^{-1}(\mathbf{u} - \mathbf{c} - \mathbf{g} - \mathbf{d}) \end{bmatrix}, \tag{37}$$

where the inverse notation communicates the mathematical mapping only; the implementation uses a linear solve.

For a long slender axisymmetric pendulum, a documented reduced model may use $J_{1r} = J_{1t} = J_1$ and $J_{1a} \simeq 0$. The full three-inertia form remains the authoritative model because dropping the transverse inertia can materially alter aggressive-motion dynamics.

6.1 Mass-matrix invertibility

The matrix is symmetric. Let $s = \sin\theta_1$ and $c = \cos\theta_1$. Since $D = J_{1r} + m_1 l_c^2 > 0$, positive definiteness follows if the Schur complement is positive:

$$A - \frac{B^2 c^2}{D} = J_0 + J_{1a} c^2 + (m_1 l_c^2 + J_{1t}) s^2 \tag{38}$$

$$+ m_1 L_0^2 \frac{J_{1r} + m_1 l_c^2 s^2}{D} > 0. \tag{39}$$

Thus $\det \mathbf{M} = D(A - B^2 c^2/D) > 0$ for positive nondegenerate mass, length, and transverse/hinge inertia with $J_0 > 0$ and nonnegative axial inertia. The synthetic set's minimum sampled determinant was 1.823076×10^{-5} and minimum eigenvalue was 1.858008×10^{-3} over 2000 angles.

7 Actuation, transmission, and friction boundary

The present input τ is torque delivered at the arm axis after the motor, gear train, and drive electronics. If motor current i becomes a state, a consistent extension is

$$\tau = \eta N K_t i, \tag{40}$$

$$L_m \dot{i} = v - R_m i - K_e N \dot{\theta}_0, \tag{41}$$

for ratio N, efficiency η, torque constant K_t, back-emf constant K_e, resistance R_m, inductance L_m, and applied voltage v. Reflected rotor inertia $N^2 J_m$ belongs in J_0 only if it has not already been included in an identified effective inertia. No electrical parameters are invented in T03.

The model includes b_0 and b_1 through Equation (17). Static and Coulomb friction, stiction, gearbox dead zone/backlash, arm compliance, motor saturation, and sensor dynamics require measured data and are deferred.

8 Parameter definitions and provenance

Table 2: Authoritative parameter meanings.

Symbol	Meaning	SI unit	Status/provenance
L_0	Base vertical axis to pendulum pivot	m	Symbolic; obtain from CAD
L_p	Total physical pendulum length	m	Symbolic; not substituted for l_c
l_c	Pendulum pivot to assembled COM	m	Symbolic; obtain from CAD/balance
m_1	Assembled pendulum mass	kg	Symbolic; weigh assembly

Symbol	Meaning	SI unit	Status/provenance
J_0	Effective base/arm inertia about $\mathbf{e}_z$, excluding pendulum	kg m^2	Symbolic; document bare arm, attachments, and reflected rotor separately
J_{1r}	Pendulum COM inertia about radial hinge axis	kg m^2	Symbolic; CAD/identification
J_{1t}	Pendulum COM transverse inertia in swing plane	kg m^2	Symbolic; CAD/identification
J_{1a}	Pendulum COM longitudinal inertia	kg m^2	Symbolic; CAD/identification
b_0, b_1	Arm and pendulum viscous damping	N m s/rad	Symbolic; identify experimentally
g	Gravitational acceleration	m/s^2	Literature; 9.81 in validation
τ	Effective mechanical arm-axis torque	N m	Model input

The JSON validation registry attaches a value, unit, description, provenance, and status to every numerical parameter. Its values are deliberately labeled `synthetic_validation`; none may be copied into the hardware model as a measurement. The brief's I_0/J_0 difference is treated as notation only after the physical axis and reference point have been confirmed.

9 Limiting cases and physical interpretation

9.1 Equilibria and gravity sign

With zero velocities and input, $\theta_1 = 0$ and $\theta_1 = \pi$ are rest configurations for any constant θ_0, because θ_0 is cyclic. Near upright, $\theta_1 = \varepsilon$ gives $g_2 \simeq -m_1 g l_c \varepsilon$, a destabilizing negative stiffness in Equation (31). Near downward, $\theta_1 = \pi + \varphi$ gives $g_2 \simeq +m_1 g l_c \varphi$, which is restoring.

9.2 Clamped-arm limit

Setting $\dot{\theta}_0 = \ddot{\theta}_0 = 0$ gives the physical pendulum equation

$$D\ddot{\theta}_1 + b_1\dot{\theta}_1 - m_1 g l_c \sin\theta_1 = 0. \tag{42}$$

The arm constraint requires the reaction/holding torque

$$\tau_{\text{hold}} = B\cos\theta_1\ddot{\theta}_1 - B\sin\theta_1\dot{\theta}_1^2. \tag{43}$$

Therefore "clamped arm" must not be combined with the assumption $\tau = 0$.

9.3 Energy balance

The total mechanical energy is $E = T + V$. Direct symbolic differentiation and contraction of Equation (32) with $\dot{\mathbf{q}}$ gives

$$\dot{E} = \tau\dot{\theta}_0 - b_0\dot{\theta}_0^2 - b_1\dot{\theta}_1^2. \tag{44}$$

This exact identity is independently simplified to zero in `src/derive_symbolic.py`. It supplies a strong sign and implementation check: unforced conservative trajectories preserve E, while unforced damped trajectories cannot gain energy.

10 Independent validation

The executable acceptance suite uses symbolic expressions derived from the kinematics and energies independently of the numerical matrix assembly. Table 3 records the actual run results. V10 is an environment status and is non-blocking. V11 records the final PDF compile and page-image inspection.

Table 3: Validation results from the machine-readable validation record.

ID	Check	Status	Recorded result
V01	Independent Euler–Lagrange agreement	PASS	Both equation differences simplify exactly to $[0, 0]^T$.
V02	Dimensional consistency	PASS	Energy terms are joules; all equation terms are N m.
V03	Mass matrix	PASS	Symbolically symmetric; analytic proof in Equation (39); sampled eigenvalues and determinants positive.
V04	Equilibria/gravity	PASS	Upright coefficient -1.648080×10^{-1}; downward coefficient $+1.648080 \times 10^{-1}$ N m/rad.
V05	Energy identity	PASS	Symbolic power-residual identity simplifies exactly to zero.
V06	Clamped arm	PASS	Physical pendulum equation and nonzero holding torque recovered.
V07	Random numerical algebra	PASS	5000 samples; maximum normalized residual 3.833679×10^{-16}; minimum eigenvalue 1.858008×10^{-3}.
V08	Conservative RK4	PASS	5 s relative maximum energy drift: 8.543×10^{-6} at 0.01 s, 3.004×10^{-7} at 0.005 s, 1.566×10^{-8} at 0.0025 s.
V09	Damped RK4	PASS	Energy fell from 0.3184634 J to 0.01045345 J; largest increment -7.705×10^{-9} J; relative power mismatch 1.047×10^{-7}.
V10	MATLAB R2024b	NOT RUN	Executable found, but MathWorks licensing services returned error 5201 with a writable temporary preferences directory.
V11	PDF compile/render QA	PASS	Eleven pages rendered at 120 dpi and inspected; no clipping, overlap, broken references, or missing glyphs.

11 Cross-check against the published full model

Cazzolato and Prime derive the full Furuta dynamics independently by Lagrange and Newton–Euler methods, retain all three pendulum principal inertias, and discuss the slender-axisymmetric reduction. Their notation uses Arm 1 for the horizontal member, Arm 2 for the pendulum, and sets the pendulum angle to zero at the stable downward equilibrium. This report instead sets the pendulum angle to zero upright.

A convention-safe mapping is

$$\theta_1^{\mathrm{CP}} = \theta_0, \qquad \theta_2^{\mathrm{CP}} = \sigma(\theta_1 - \pi), \qquad L_1^{\mathrm{CP}} = L_0, \qquad l_2^{\mathrm{CP}} = l_c, \tag{45}$$

where $\sigma = +1$ when the paper's front-view positive direction is chosen to match this report's swing-plane positive direction and $\sigma = -1$ when the opposite viewing side is used. Associated off-diagonal and generalized-force signs transform with the same σ; physical predictions are unchanged. The paper's Arm 2 mass maps to m_1 here, while inertia subscripts must be mapped by physical principal axis rather than by letter alone. The horizontal arm pivot inertia maps to this report's effective J_0 only after parallel-axis, attachment, and motor-reflection contributions are reconciled. Under these translations, the full-model structure—configuration-dependent arm inertia, cos off-diagonal coupling, centrifugal/Coriolis terms, gravity, and viscous damping—agrees. This is a structural literature cross-check; V01 is the independent algebraic proof for the exact convention used here.

12 Limitations and next-task handoff

- CAD geometry and inertias are not yet confirmed. Replace the synthetic registry only with traceable measurements or CAD exports.
- The model assumes diagonal COM inertia. If CAD reports products of inertia, rotate the tensor into the adopted body frame before simplifying.
- Motor voltage/current, saturation, friction nonlinearities, backlash, and flexibility are outside T03.
- MATLAB-compatible files are supplied, but the installed MATLAB R2024b could not be executed because its licensing service was unavailable.
- T04 should linearize the supplied state derivative about $\theta_1 = 0$, zero rates, zero torque, and any fixed θ_0.
- A Simscape or CAD comparison must first reproduce Figure 1; otherwise sign agreement is not meaningful.

13 Conclusion

The T03 nonlinear mechanical model is complete and internally consistent. It retains the inertia terms needed for a full Furuta model, exposes the common slender-pendulum reduction without silently imposing it, and separates mathematical validation from hardware parameter identification. The exact symbolic identities, analytic mass-matrix proof, and energy-based numerical tests provide a reproducible baseline for linearization, control design, and multibody-model comparison.

References

1. B. S. Cazzolato and Z. Prime, "On the Dynamics of the Furuta Pendulum," *Journal of Control Science and Engineering*, vol. 2011, Article ID 528341, 2011. https://doi.org/10.1155/2011/528341.